A positive moment measure for the Fennessey–Larcombe–French determinant sequence

Principia Math In Progress

11 September 2026

Abstract

We prove Conjecture 5.1 of Sun and Zhao: the sequence $V_n^2 - V_{n-1}V_{n+1}$, $n \geq 2$, is infinitely log-convex. In fact it is a Stieltjes moment sequence with an explicit positive integral density after normalization. Positivity is proved analytically on two endpoint intervals and by four rigorous, finite rational certificates on the remaining interval. The certificate uses no floating-point arithmetic.

Task 1: Resolve MathDB problem 332482

Let $V_0 = 1$, $V_1 = 8$, and, for $n \geq 2$, let

$$(n-1)n^2V_n = 8(n-1)(3n^2 - n - 1)V_{n-1} - 128(n-2)n^2V_{n-2}. \tag{1}$$

Set $D_n = V_n^2 - V_{n-1}V_{n+1}$ for $n \geq 2$. For a sequence $(a_m)_{m\geq 0}$ define $(\mathcal{L}a)_m = a_m a_{m+2} - a_{m+1}^2$. Infinite log-convexity means that every iterate $\mathcal{L}^r a$ is nonnegative. This is the usual convention with its initial index shifted at each iteration.

Theorem 1. *The sequence $(D_{m+2})_{m\geq 0}$ is a Stieltjes moment sequence represented by a finite positive measure on $[0, 256]$ with support equal to that interval. It is strictly infinitely log-convex. In particular, published Conjecture 5.1 [1, p. 601] holds.*

The MathDB title says Catalan–Larcombe–French, but its statement and the published conjecture both specify the Fennessey–Larcombe–French numbers V_n used here.

A preliminary density identity

We give the needed representation explicitly, including sequence identification. Write

$$L = \frac{2}{\pi}, \quad M = 2L - 1, \quad c_j = \left(\frac{\binom{2j}{j}}{4^j}\right)^2, \quad K(m) = \int_0^{\pi/2} (1 - m\sin^2\theta)^{-1/2}\, d\theta.$$

Thus K uses the parameter convention, not the modulus convention. For $u \neq 1/2$ in $(0, 1)$ set

$$w(u) = \frac{4}{\pi^2} K(4u(1-u)), \qquad \rho(u) = w(u) + (2u-1)w'(u). \tag{2}$$

Values at isolated singularities are irrelevant to all measures below.

Lemma 1. *The density ρ is positive almost everywhere, symmetric about $1/2$, and has mass $M < L$. For $n \geq 1$,*

$$v_n := \frac{V_n}{16^n} = L - \int_0^1 u^n \rho(u)\, du. \tag{3}$$

It has the positive series

$$\rho(u) = \frac{2}{\pi} \sum_{j \geq 1} \frac{j}{j+1} c_j [4u(1-u)]^j. \tag{4}$$

Proof. Let X, Y be independent with density $1/(\pi\sqrt{1-x^2})$ on $(-1,1)$, and put $U = (1+XY)/2$. The product-density substitution gives

$$f_{XY}(t) = \frac{2}{\pi^2} \int_{|t|}^1 \frac{dx}{\sqrt{(1-x^2)(x^2-t^2)}} = \frac{2}{\pi^2} K(1-t^2).$$

The second equality follows by setting $x^2 = t^2 + (1-t^2)\sin^2\theta$ and reversing the angle. Consequently w is the probability density of U and

$$p_n := \mathbb{E}(U^n) = 2^{-n} \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j} c_j.$$

Here the arcsine even moments are $\binom{2j}{j}/4^j$. To verify the recurrence identification, let $H(q) = \sum c_j q^j$. The coefficient ratio gives $q(1-q)H'' + (1-2q)H' - H/4 = 0$. The generating function of p_n is $A(z) = 2(2-z)^{-1} H(z^2/(2-z)^2)$, whence, by substitution,

$$z(1-z)(2-z)A'' + (2 - 6z + 3z^2)A' + (z-1)A = 0.$$

Thus $P_n = 16^n p_n$ satisfies, for $n \geq 2$,

$$E_n := n^2 P_n - 8(3n^2 - 3n + 1)P_{n-1} + 128(n-1)^2 P_{n-2} = 0,$$

with $P_0 = 1, P_1 = 8$ and $E_1 = P_1 - 8P_0 = 0$. Set $W_n = (2n+1)P_n - 16nP_{n-1}$ for $n \geq 1$. The residual of (1) with W in place of V is $(n-1)(2n+1)E_n - 16n^2 E_{n-1}$. It vanishes, and $W_1 = 8$, so $W_n = V_n$. In particular,

$$v_n = \int_0^1 \left[u^n + n(2u-1)u^{n-1}\right] w(u)\, du. \tag{5}$$

Expanding $K(m) = (\pi/2)\sum c_j m^j$ in (2) gives (4); indeed $\rho = (4/\pi^2)[K(m) - 4(1-m)K'(m)]$ with $m = 4u(1-u)$. The series proves positivity and symmetry. The elementary bound $c_j \leq 1/(j+1)$, obtained by induction from the coefficient ratio, gives $K(m) = O(1 + |\log(1-m)|)$. Hence $h(u) = (2u-1)w(u)$ extends continuously through $1/2$ with value zero. Its endpoint values are $h(0) = -L, h(1) = L$, and away from $1/2$ its derivative is $h' = w + \rho > 0$. Integrating on each side and taking limits gives $\int_0^1 (w + \rho) = 2L$. Thus ρ is integrable of mass $M = 2L - 1$, and h is absolutely continuous. Integration by parts in (5) now gives (3). The boundary contribution at zero is zero because $n \geq 1$. Finally $M < L$ since $L < 1$. □

The determinant density

For almost every $0 < t < 1$ define

$$A_0(t) = L(1-t)^2\rho(t), \tag{6}$$

$$N(t) = \frac{1}{2}\int_t^1 (s - t/s)^2\rho(s)\rho(t/s)\,\frac{ds}{s}, \qquad d(t) = A_0(t) - N(t). \tag{7}$$

Both A_0 and N are integrable and nonnegative: by Tonelli, $\int N = \frac{1}{2}\int\int (s-u)^2\rho(s)\rho(u)\,ds\,du \le M^2/2$. Expanding (3) and changing variables $t = su$ gives

$$\frac{D_n}{256^n} = v_n^2 - v_{n-1}v_{n+1} = \int_0^1 t^{n-1}d(t)\,dt \qquad (n \ge 2). \tag{8}$$

For clarity, if $a_k = \int u^k\rho(u)\,du$, the left side is $L(a_{n-1} - 2a_n + a_{n+1}) - (a_{n-1}a_{n+1} - a_n^2)$; the latter parenthesis is $\frac{1}{2}\int\int (su)^{n-1}(s-u)^2\rho(s)\rho(u)\,ds\,du$. All changes of variables therefore involve finite nonnegative integrals. The symmetry $s \leftrightarrow t/s$ also gives

$$N(t) = \int_{\sqrt{t}}^1 \frac{(a - t/a)^2}{a}\rho(a)\rho(t/a)\,da. \tag{9}$$

We will prove $d(t) > 0$ almost everywhere.

Positivity for $1/2 \le t < 1$

The series (4) shows that ρ decreases on $(1/2, 1)$. For $\sqrt{t} \le a \le 1$, put $b = t/a$. Then $t \le b \le a$ and $\rho(b) \le \rho(t)$. Moreover $a - b \le a(1-t)$, so $(a-b)^2/a \le (1-t)^2$. Consequently (9) yields

$$N(t) \le M(1-t)^2\rho(t) < L(1-t)^2\rho(t) = A_0(t). \tag{10}$$

The value at $t = 1/2$ may be omitted.

Positivity for $0 < t \le 1/4$

For $0 < u < 1/2$, the descending Landen identity [2, 19.8.11–12] in parameter convention is

$$K(4u(1-u)) = \frac{1}{1-u}K\left(\frac{u^2}{(1-u)^2}\right).$$

In DLMF's modulus convention use $k = 2\sqrt{u(1-u)}$, $k' = 1-2u$, and $k_1 = u/(1-u)$. Differentiation and the series for K give

$$\rho(u) = \frac{u}{\pi(1-u)^2}G\left(\frac{u^2}{(1-u)^2}\right), \qquad G(z) = \sum_{j\ge 0}\frac{2j+1}{j+1}c_j z^j. \tag{11}$$

For example, the coefficient calculation uses $(1+2j)c_j - 2(j+1)c_{j+1} = (2j+1)c_j/(2(j+1))$. Expanding the last factor in the following beta integral proves

$$G(z) = \frac{2}{\pi}\int_0^1 \frac{r^{1/2}\,dr}{\sqrt{1-r}\sqrt{1-zr}}.$$

It follows pointwise under the integral that

$$\frac{G(z_1)}{G(z_0)} \le \sqrt{\frac{1-z_0}{1-z_1}} \qquad (0 \le z_0 \le z_1 < 1). \tag{12}$$

We claim that whenever $0 < b \le a \le 1$ and $ab \le 1/4$,

$$\frac{(a-b)^2}{a}\rho(b) \le (1-ab)^2\rho(ab). \tag{13}$$

Except at the irrelevant coincident midpoint, $b < 1/2$. By (11)–(12), the ratio of the left side to the right side is at most

$$F(a,b) = \frac{(a-b)^2}{a^2(1-b)(1-ab)}\sqrt{\frac{1-2ab}{1-2b}}.$$

For $a > b$, its logarithmic derivative in a is

$$\partial_a \log F = \frac{2b}{a(a-b)} - \frac{ab^2}{(1-ab)(1-2ab)} > 0.$$

Indeed, its sign is that of $2 - 6t + 5t^2 - a^2 t$, where $t = ab$, and this is at least $2 - 7t + 5t^2 = (1-t)(2-5t) > 0$ on the required range. If $b \le 1/4$, increasing a to 1 gives $F(a,b) \le F(1,b) = 1$. If $1/4 < b < 1/2$, increase a to $1/(4b)$ instead. There,

$$F(1/(4b), b) = \frac{4}{3\sqrt{2}}\frac{(1-2b)^{3/2}(1+2b)^2}{1-b}.$$

Its logarithmic derivative has the sign of $2 - 15b + 10b^2 < 0$ on $[1/4, 1/2)$; at $b = 1/4$ its value is 1. The case $a = b$ follows directly or by a limit. This proves (13). Substituting $b = t/a$ in (9) and integrating against $\rho(a)\,da$ proves (10) also for $0 < t \le 1/4$.

A finite certificate for $1/4 \le t < 1/2$

The Cauchy–Schwarz inequality for gamma integrals gives $\Gamma(j+1/2)^2 \le \Gamma(j)\Gamma(j+1)$, hence $c_j \le 1/(\pi j)$ for $j \ge 1$. With $m = 4s(1-s)$, (4) implies

$$\rho(s) \le \frac{2}{\pi^2}\sum_{j\ge 1}\frac{m^j}{j+1} \le -\frac{2}{\pi^2}\log(1-m) = \frac{4}{\pi^2}\big(-\log|1-2s|\big). \tag{14}$$

In (7), use $2xy \le x^2 + y^2$ and the change $s \leftrightarrow t/s$ in one of the two resulting integrals. This gives

$$N(t) \le \frac{8}{\pi^4}\int_t^1 Q(s,t)\log^2|1-2s|\,ds, \qquad Q(s,t) = \frac{(s-t/s)^2}{s}. \tag{15}$$

On the other hand (11) and $G \ge 1$ give

$$A_0(t) \ge \frac{2t}{\pi^2}. \tag{16}$$

For $j = 4, 5, 6, 7$, put $a_j = j/16$, and take $t \in [a_j, (j+1)/16]$. For $i = 4j, \dots, 63$, define the rational number

$$B_{j,i} = \max\{Q(s,t) : s \in \{i/64, (i+1)/64\},\ t \in \{j/16, (j+1)/16\}\}.$$

These four corner values bound Q on the whole rectangle. For fixed s, Q is convex in t; for fixed t,

$$\partial_s Q = \frac{(s^2 - t)(s^2 + 3t)}{s^4},$$

so its only interior critical point in $s > 0$ is a minimum. Let

$$W_i = \int_{i/64}^{(i+1)/64} \log^2 |1 - 2s| \, ds, \qquad I_j = \sum_{i=4j}^{63} B_{j,i} W_i.$$

Enlarging the nonnegative integral in (15) down to a_j shows that it is at most I_j. The exact certificate below proves

j	certified rational upper bound for I_j	$2a_j$
4	385126/1000000	1/2
5	521335/1000000	5/8
6	687221/1000000	3/4
7	805114/1000000	7/8

Every middle entry is strictly below the corresponding right entry. Since $\pi^2 > 8$, it follows that

$$N(t) < \frac{16 a_j}{\pi^4} < \frac{2a_j}{\pi^2} \le \frac{2t}{\pi^2} \le A_0(t).$$

Together with the endpoint arguments, this proves $d > 0$ almost everywhere on $(0, 1)$.

Here are all details for verifying the finite table. Put $\Phi(0) = 0$ and $\Phi(y) = y(\ell^2 + 2\ell + 2)$ for $y > 0$, where $\ell = -\log y$. Thus $\Phi'(y) = \log^2 y$. Only arguments $y = k/32$, $0 \le k \le 32$, occur in the weights. For $y > 0$, set $z = (1 - y)/(1 + y)$ and

$$\ell_- = 2 \sum_{k=0}^{127} \frac{z^{2k+1}}{2k + 1},$$
$$\ell_+ = \ell_- + \frac{2z^{257}}{257(1 - z^2)}.$$

The power series $-\log y = 2 \operatorname{arctanh} z$ and a geometric tail estimate prove $\ell_- \le \ell \le \ell_+$. The function $y(\ell^2 + 2\ell + 2)$ increases in $\ell \ge 0$. Round its lower bound down and upper bound up to multiples of 2^{-80}, obtaining $\Phi_-(y)$ and $\Phi_+(y)$. For $i < 32$ use

$$W_i \le \frac{\Phi_+((32 - i)/32) - \Phi_-((31 - i)/32)}{2};$$

for $i \ge 32$ use

$$W_i \le \frac{\Phi_+((i - 31)/32) - \Phi_-((i - 32)/32)}{2}.$$

These are rational expressions with outward bounds throughout. Multiplying by $B_{j,i}$, summing, and rounding up to the next multiple of 10^{-6} gives the displayed table. The accompanying standard-library Python program `verify_density_certificate.py` implements these exact instructions with `fractions.Fraction`; its complete rational sums are retained in `density-certificate.json`. This finite arithmetic certificate is a part of the proof.

Moment closure and all iterations

Proof of Theorem 1. Define ν to be the image under $x = 256t$ of the positive measure with density $256^2 t\, d(t)$ on $(0,1)$. It is finite and has support $[0,256]$. Equation (8) gives $D_{m+2} = \int x^m \, d\nu(x)$ for every $m \geq 0$.

For any finite positive measure η supported on $[0,B]$, if $a_m = \int x^m \, d\eta(x)$, then

$$a_m a_{m+2} - a_{m+1}^2 = \frac{1}{2} \int\int (xy)^m (x-y)^2 \, d\eta(x)\, d\eta(y).$$

This is again a moment sequence: use the image under multiplication of $(x-y)^2 \eta \otimes \eta/2$, supported on $[0, B^2]$. Iteration proves infinite log-convexity. If η has full support $[0,B]$ with $B > 0$, this new measure has full support $[0, B^2]$: every point in $(0, B^2)$ is a product of two distinct points of $(0,B)$, and small neighborhoods have positive product measure and positive weight. Thus every iterate has positive moments, proving the asserted strictness. □

Scope

The proof covers all iteration orders and all required indices. It uses the classical Landen identity, proved endpoint estimates, and a reproducible exact finite certificate. It does not infer a uniform inequality from numerical samples. The current published source and live claim fields were checked before completion; the separate literature audit specifies the search scope. No external peer review or guaranteed novelty is claimed.

Task 2: Prove the optimal global density bound

Determine the least constant C for which $N(t) \leq C(1-t)^2 \rho(t)$ almost everywhere on $(0,1)$. In particular, prove or refute the candidate $C = M = 4/\pi - 1$, which the endpoint argument already establishes outside $(1/4, 1/2)$. Seek an analytic argument that also removes the finite certificate.

Task 3: Derive quantitative bounds for all iterates

Use the explicit moment measures to bound $(\mathcal{L}^r(D_{m+2}))_m$ above and below in terms of m and r. Obtain sharp asymptotics for fixed r as $m \to \infty$, then determine how far those estimates remain uniform when r grows with m.

Task 4: Classify shifted determinant densities

For each integer $h \geq 2$, analyze $V_n^2 - V_{n-h} V_{n+h}$ for $n \geq h+1$. Derive its analogue of $d(t)$ and determine which h yield a Stieltjes moment sequence after shifting the initial index. Identify any failure by a rigorous obstruction or explicit counterexample.

Task 5: Extend the measure construction to beta laws

Replace the arcsine variables in the preliminary lemma by independent densities proportional to $(1-x^2)^{a-1}$ and $(1-y^2)^{b-1}$, where $a, b > 0$. With $U = (1+XY)/2$ and $B_n = (2n+1)\mathbb{E}(U^n) - n\mathbb{E}(U^{n-1})$, determine the parameter region where the determinant sequence $B_n^2 - B_{n-1} B_{n+1}$ is a positive moment sequence. Include the changed endpoint contributions.

References

[1] B. Y. Sun and J. J. Y. Zhao, *Log-behavior of Two Sequences Related to the Elliptic Integrals*, Acta Mathematicae Applicatae Sinica, English Series **36** (2020), 590–602, Conjecture 5.1, p. 601. https://doi.org/10.1007/s10255-020-0952-1. Full publisher PDF: https://applmath.cjoe.ac.cn/jweb_yysxxb_en/EN/PDF/10.1007/s10255-020-0952-1.

[2] NIST Digital Library of Mathematical Functions, Section 19.8(ii), descending Landen transformation, equations 19.8.11–12. https://dlmf.nist.gov/19.8#ii.